



GRiD 8.0

PROMPT THE FUTURE

Flipkart's journey began in 2007 with a bold leap of faith, evolving from a startup into the backbone of Indian e-commerce. From pioneering Cash on Delivery to architecting the massive infrastructure behind Big Billion Days and launching quick commerce in 2024, our path has been defined by a relentless bias for action. We didn't just wait for the ecosystem to mature; we built the roads, the supply chains, and the hyper-scale systems required to bring digital commerce to millions.

At the center of this evolution is an unwavering commitment to customer-centricity. True innovation isn't just about writing complex code; it is about solving deep-rooted, real-world problems for a diverse nation. We view technology not as a static tool, but as an intelligent engine designed to empower every Indian's dream—delivering value through cutting-edge tech and engineering at scale.

As we move into the AI era, our systems are being re-engineered around complex human needs. GRiD 8.0 is our invitation for you to continue our legacy, & 'prompt the future' with us!



Track: Software Development

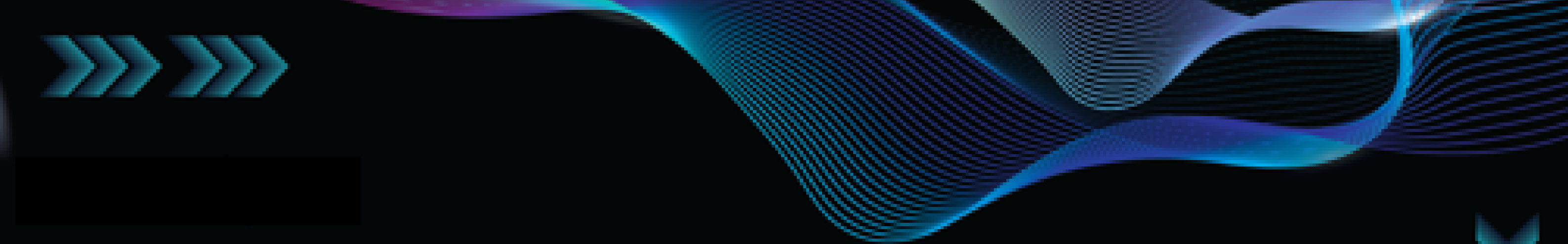
Context

In the competitive landscape of e-commerce, cart abandonment acts as a silent profit killer. Customers frequently load their carts only to depart without converting, driven by a spectrum of friction points—from uncertainty about product quality or delivery timelines to pricing sensitivity, payment hurdles, or simply loss of interest.

Conventional industry approaches, such as distributing uniform, blanket coupons or generic reminders, frequently fall flat. These methods fail because they ignore the customer's specific intent and the unique, underlying cause of their hesitation.

The status quo user journey for a potential drop-off currently looks like this:

1. **Session Start:** Customer browses, adds items to cart.
2. **The Hesitation:** User exhibits pre-abandonment behaviors (e.g., repeatedly viewing reviews, bouncing between cart and product page).
3. **The Missed Opportunity:** The system triggers a standard, non-personalized reminder (or nothing at all).
4. **Abandonment:** Customer leaves the platform.



Core Business Challenge

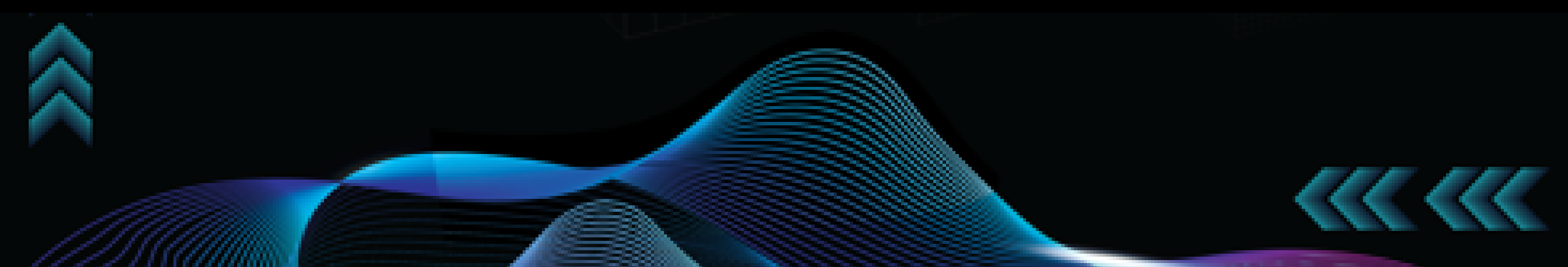
Currently, using rigid recommendations to treat this problem has led to some friction points:

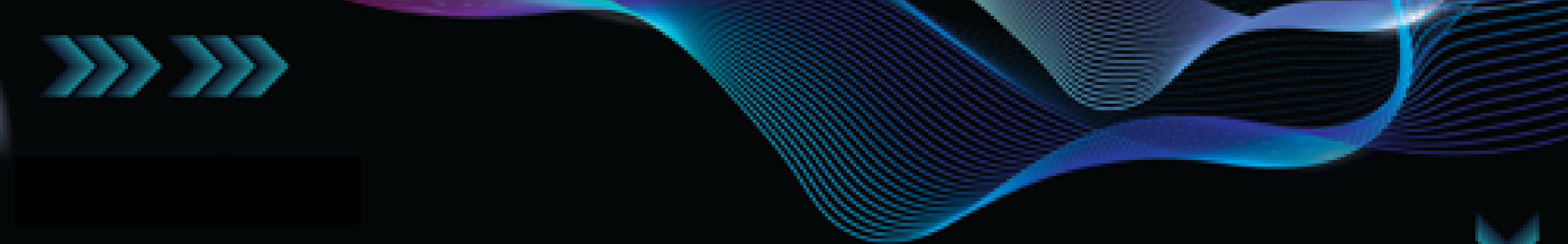
- **Intent Blindness:** Treating all abandoning users as a monolith leads to irrelevant interventions (e.g., offering a discount to a user who is actually concerned about shipping times).
- **Conversion Drag:** Missing the precise "window of influence" means we lose a customer who might have converted with just a nudge.
- **Margin Erosion:** Giving blanket discounts to users who would have converted anyway wastes promotional budget.

The core business challenge is transitioning from these rigid, rule-based reminders to an Intelligent Agentic System. We need a solution that mimics the logic of a helpful shopping assistant—assessing context, historical behavior, and real-time activity to provide the exact intervention required to bridge the gap to purchase. By automating personalized interventions, we maximize conversion rates while maintaining healthy margins.

Problem Statement

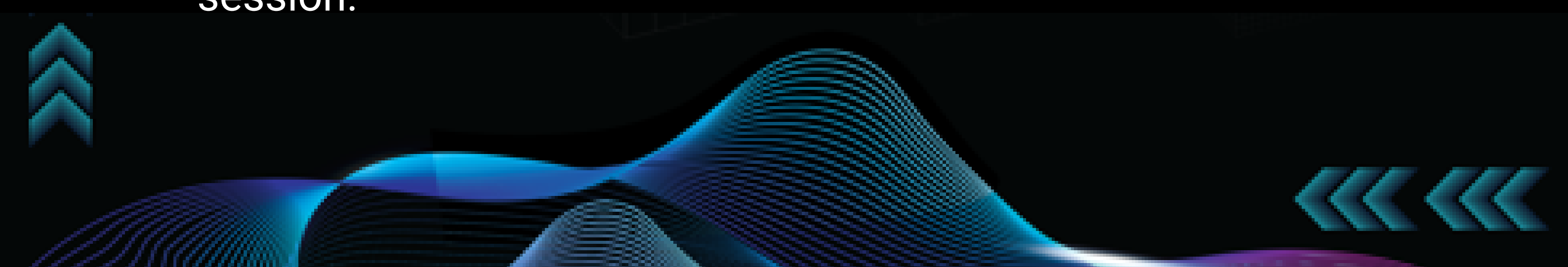
You are tasked with engineering an intelligent system designed to autonomously predict cart abandonment and deploy precise, personalized interventions. Your solution must bridge the gap between static rule-based reminders and intent-aware, human-like reasoning.

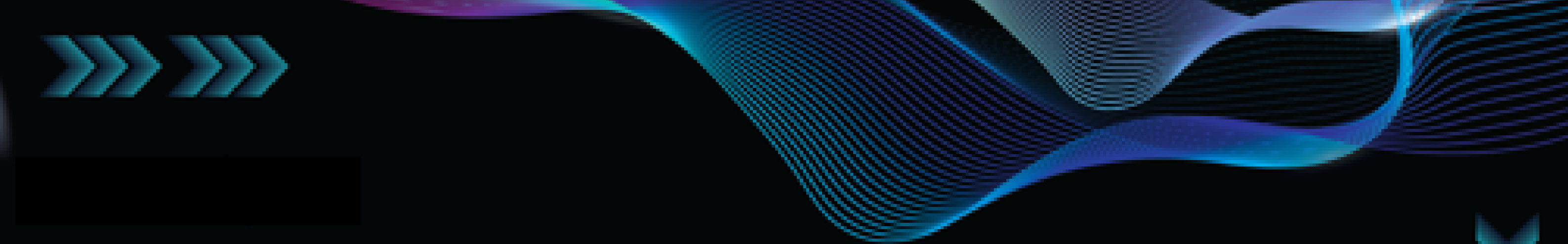




The system must consume a multifaceted data input package: purchase history, cart metadata, session duration, search patterns, and device signals. Your objective is not just to output a generic nudge, but to function as a sophisticated decision-making agent. It must evaluate these inputs to provide a final output: Abandonment Probability, The Root Cause, The Recommended Intervention, and a Confidence Score.

Your system must explain why a particular intervention was recommended. If the agent is unsure, it should default to a non-intrusive action rather than a high-cost discount. Technical challenges to keep in mind:

- **Multi-Agent Orchestration:** You cannot rely on a single heuristic. You must orchestrate specialized agents (e.g., a Feature Agent for real-time behavior retrieval, a Reasoning Agent for intent analysis, and a Recommendation Agent for intervention selection).
 - **Explainability:** An opaque model is counter-productive. Your system must produce a human-readable justification trail that connects the evidence (e.g., "User viewed similar products 8 times" + "revisited reviews" = "Product Quality Uncertainty") to the specific intervention.
 - **Personalization vs. Over-Serving:** Balance the tradeoff between conversion (pushing an intervention) and user experience (avoiding "spammy" behavior).
 - **Real-time Processing:** The system must evaluate the risk and deploy the intervention in real-time, before the user exits the session.
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Examples of Interventions

- Delivery reassurance
- Product comparison
- Price-drop notification
- Similar product recommendation
- Limited-time discount
- EMI suggestion
- Product reviews summary
- Wishlist reminder, etc.

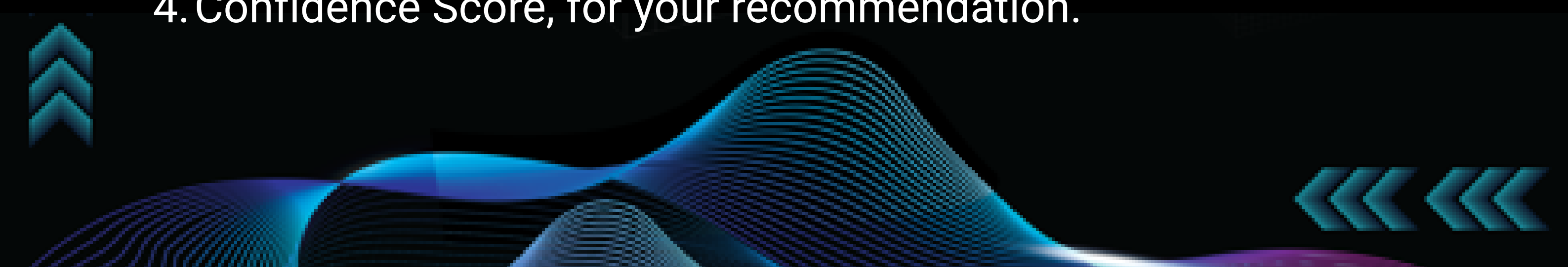
Sample Input

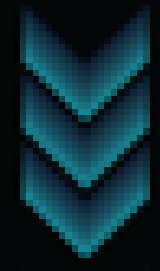
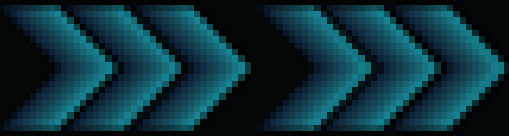
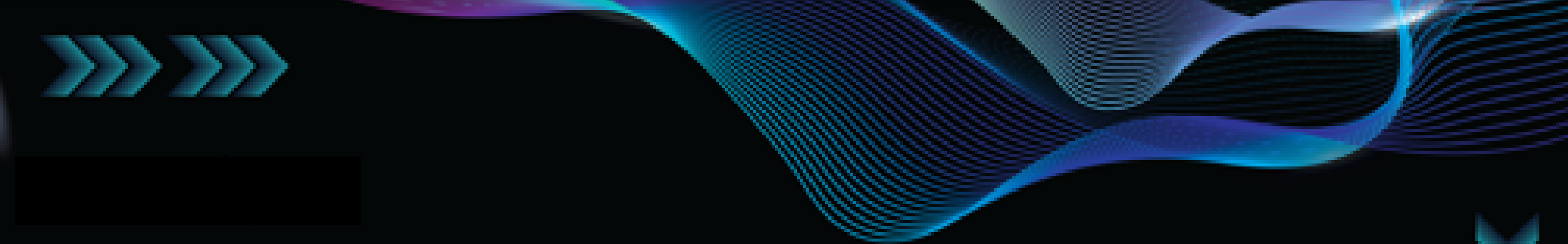
Each cart session may contain information such as: Purchase history, Previous cart abandonment, Product category, Cart value, Number of products, Session duration, Number of searches, Product page visits, Delivery availability, Price changes, Wishlist history, Device type, Time of day, User demographics (optional).

Participants can choose a subset of above inputs OR can add more inputs and create synthetic datasets accordingly.

Expected Output

For each active cart:

1. **Abandonment Probability:** How likely is it for the customer to abandon the cart?
 2. **Root Cause:** What is the most likely reason for the customer to abandon the cart?
 3. **Recommended Intervention:** Solution & reason for the same.
 4. Confidence Score, for your recommendation.
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Sample Output As Per Expectations

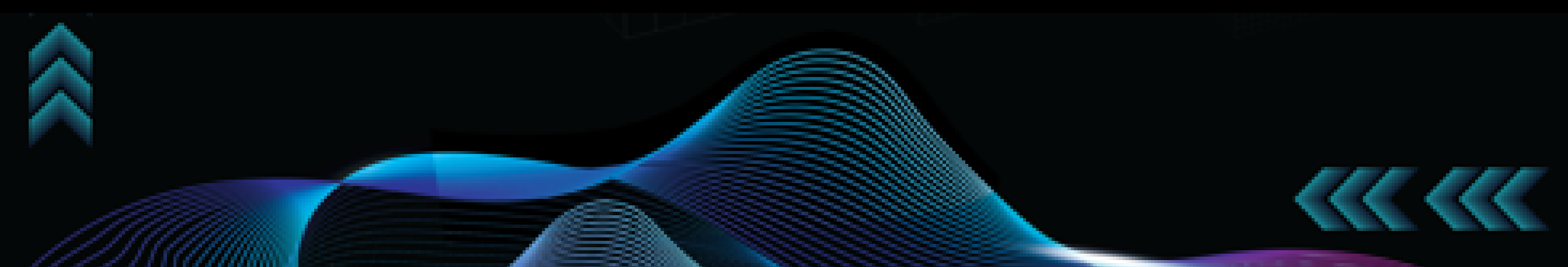
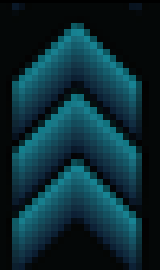
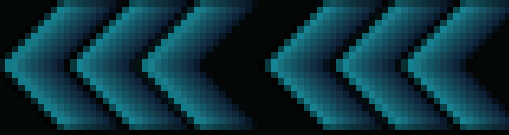
- **Abandonment Probability** : 82%
- **Likely Reason**: Customer is comparing prices and uncertain about product quality.
- **Explanation**: User viewed similar products 8 times and revisited reviews before leaving.
- **Recommended Intervention**: Display summarized customer reviews and offer Price Drop Alert.
- **Confidence score**: 91%

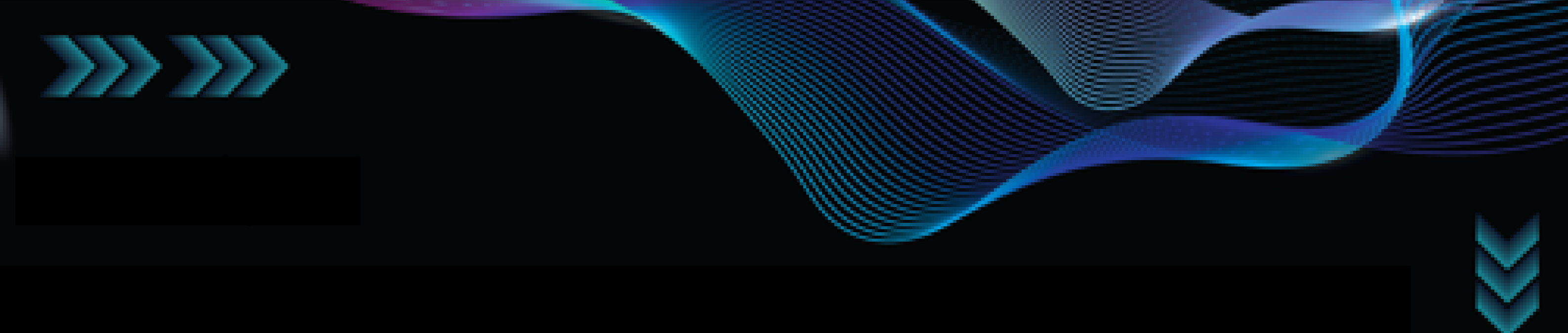
AI Opportunities

Participants may use: Classification models, Gradient Boosting, Random Forest, Deep Learning, LLMs for explanation generation, Embedding-based recommendation, RAG for product review summarization.

Deliverables

Participants should provide:

- Prediction model
 - Recommendation engine
 - Explainability module
 - Dashboard/UI
 - Architecture diagram
 - Demo
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Bonus Ideas

Participants can innovate and provide solutions over and above the defined problem statement to get additional points like:

- Multilingual recommendations
- A/B testing simulation
- Feedback loop for intervention effectiveness, etc.

Evaluation Criteria

- **Prediction Accuracy:** This measures the system's effectiveness in correctly identifying the likelihood of cart abandonment.
 - **Quality of Intervention:** This assesses the relevance and precision of the recommended solution provided by the system. The goal is to move beyond generic, blanket interventions and instead deploy context-aware nudges that directly address the root cause of hesitation.
 - **Explainability:** This requires the system to provide a human-readable justification trail that links specific evidence to the suggested intervention. It helps to build trust in the automated logic.
 - **Innovation:** This evaluates the creativity and advanced nature of the solution provided.
 - **User Experience:** This weighs how effectively the intervention is helpful and seamless rather than intrusive, ensuring it acts as a supportive shopping assistant, without being viewed as 'spammy'.
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